

BLE VITALS SCANNER

Orbytring Brand Edition

Presenter: Kavitha R

Email: kavitharamesh1408@gmail.com

Applied Position: Flutter Intern

Target Assignment: BLE Vitals Scanner (sensio / orbytring)

The Challenge & Objective

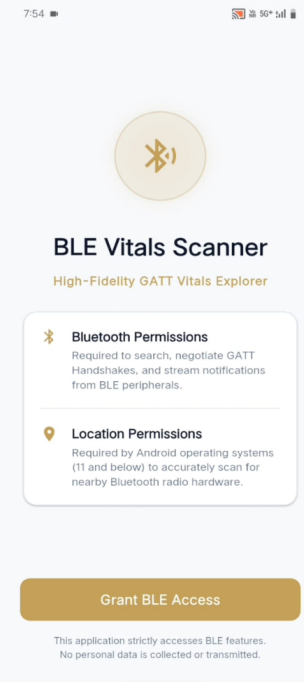
TECHNICAL MISSION STATEMENT

- **Objective:** Build a responsive, crash-free Flutter application that dynamically scans for nearby BLE signals, connects to a selected target device, and visualizes live continuous telemetry.
- **Core Constraint:** Rely exclusively on `flutter_reactive_ble` for native Bluetooth channels, permissions verification, and characteristic notification streams.
- **Design Strategy:** Reject default UI templates in favor of a brand-centric **White Theme** modeled after orbytring.com. Utilizes champagne gold and rose gold accents over metallic slate surfaces.
- **Reliability:** Verify and package a tree-shaken release build (APK) that parses raw vital streams correctly and runs warning-free.

Permissions Gate & Onboarding

DYNAMIC ONBOARDING & VALIDATION VIEW

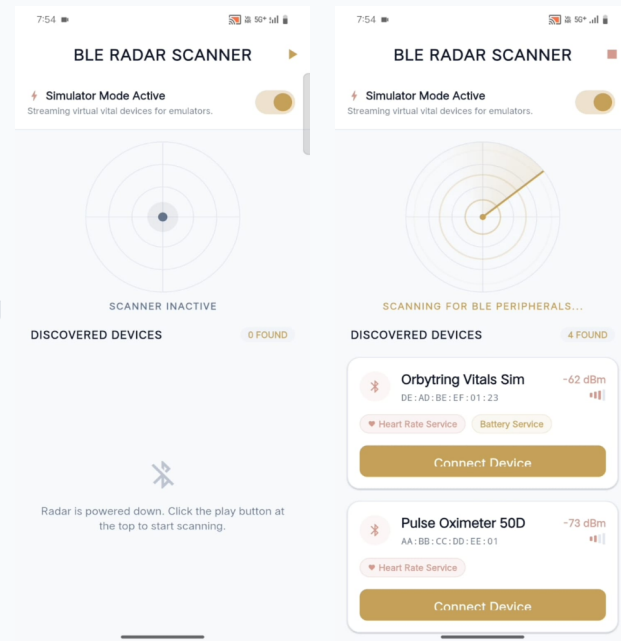
- **Cross-SDK Permissions:** Solicits appropriate Bluetooth scanner and advertiser channels dynamically based on the Android operating system level.
- **Location Checks:** Requests location permissions on SDK 30 and below, while bypassing it on newer SDKs to preserve user privacy.
- **Dynamic Buttons:** Shows real-time request states, transitioning from a CTA button 'Grant BLE Access' to loading states during handshakes.
- **Safety First:** Explains details clearly with dedicated card descriptions to maximize onboard conversions.



Visual sweeping Radar Scanner

ANIMATED SWEEPER RADAR & DYNAMIC SCANS

- **Custom Radar sweeps:** A canvas-based custom painter renders a continuous sweeping gold radar graphic highlighting search paths.
- **Simulator Swaps:** A card switch enables immediate local emulation of virtual hardware, bypassing emulator BLE limitations.
- **Signal Strength bars:** Shows discrete rssi meters and decibel readings, sorting transceivers in real-time.
- **Advertised Services:** Shows tags representing advertised services, including 'Heart Rate Service' and 'Battery Service'.



Curated Brand Design System

PREMIUM ORBYTRING BRAND TOKENS

- **Canvas Background:** Hex **#F8FAFC** (Metallic Soft White background providing clinical cleanliness).
- **Surface Elements:** Hex **#FFFFFF** (Pure White card layouts lined with metallic silver borders).
- **Auric Accent Gold:** Hex **#C5A059** (Signature Orbyt Gold branding for radar sweeps, toggles, buttons, and status icons).
- **Vitals Rose Gold:** Hex **#D49A8F** (Orbyt Rose Gold for scrolling ECG lines, nodes, and pulsing heart visualizations).
- **Slate Typography:** Hex **#0F172A** (Stellar Black primary headers) & **#64748B** (Gunmetal Grey secondary parameter data).

Robust Architecture Blueprint

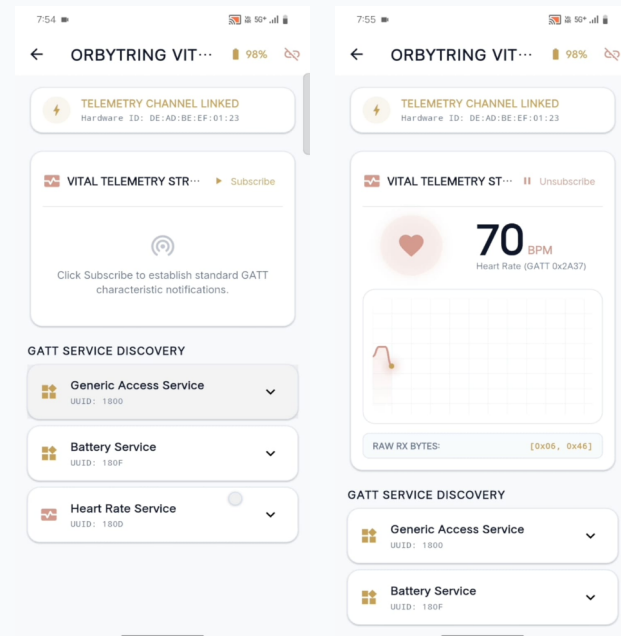
DECOUPLED SERVICE REPOSITORY DESIGN

- **BleService Interface:** Creates an abstract contract defining all required BLE properties, removing direct hardware vendor ties.
- **ReactiveBleService:** Concrete implementation powering the native radio transceiver using the *flutter_reactive_ble* stream API.
- **MockBleService:** Complete offline simulation replacement generating RSSI fluctuations, connection states, and realistic Heart Rate data.
- **Flexible Providers:** State architecture leverages *BleProvider* to manage scanning and simulator switches, while *VitalsProvider* listens to live characteristic notifications.

Clinical Vitals Dashboard

GATT EXPLORER & TELEMETRY DECK

- **Telemetry Link:** The connection status bar shows instant handshake status with hardware identifiers.
- **Continuous ECG Graph:** Plots incoming Beats Per Minute metrics on a scrolling Bezier grid.
- **Pulsing Vitals Indicator:** A dynamic favorite icon heart pulses at the exact speed of the streamed BPM values.
- **Raw Console Display:** Features a raw RX byte logger translating notifications into readable hex codes (e.g. `[0x06, 0x46]`).



GATT Specification Vitals Parsing

COMPLIANT BINARY TELEMETRY EXTRACTION

Extracts live Beats Per Minute directly from standard raw binary packets (GATT 2A37):

```
// Raw heart rate value is parsed by inspecting flag bits:
if (packet.isEmpty) {
    final int flags = packet[0];
    final int bpm;

    if ((flags & 0x01) == 0) {
        // 8-bit Heart Rate value format
        bpm = packet[1];
    } else {
        // 16-bit Heart Rate value format (little-endian representation)
        bpm = packet[1] | (packet[2] << 8);
    }
    _updateVitals(bpm);
}
```


Rigorous Code Verification

COMPILER INTEGRITY & BUILD VALIDATION

- **Static Analyzer:** Executed *flutter analyze* reporting **No issues found!** with zero errors, zero warnings, and zero lints.
- **Widget Smoke Test:** Includes dynamic layout checks that test initially mapped screens.
- **Production Release Build:** Compiled successfully using *flutter build apk --release*.
- **Delivery Output:** Highly optimized, tree-shaken Android APK compiled at **build\app\outputs\flutter-apk\app-release.apk** (44.2MB).

THANK YOU

BLE Vitals Scanner Submission

Submitted by: Kavitha R

Email Contact: kavitharamesh1408@gmail.com

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